SRM UNIVERSITY – *AP*, ANDHRA PRADESH

SRMAP / Reg. Off / Policies / 12 / 2023-24 **20th October 2023**

CENTRAL INSTRUMENTATION FACILITY (CIF) POLICY OF SRM UNIVERSITY – *AP*

The Central Instrumentation Facility at SRM University Andhra Pradesh (SRMAP) supports an impressive array of advanced instruments, which catalyse advancing research in science and technology to the new heights. These cutting-edge tools and resources empower faculty members, research scholars, and students to engage in globally competitive research across various domains, including basic, applied, and translation research.

The facility provides a modern suite of advanced instruments that offer a wide spectrum of techniques and methods for conducting chemical and material testing and analysis. As a result, researchers can anticipate producing research with high impact through the publications in peer-reviewed journals, patents, prototypes and products. Ultimately, the collective endeavours of this facility contribute to the betterment of society at large. The CIF is envisioned to be a prominent hub for pioneering and collaborative research in the country.

CIF will operate under the office of Dean Research and is expected to be self-sustaining through revenue generation for instrument upkeep and maintenance.

Sophisticated instruments under this facility can be utilized by undergraduate and postgraduate students, doctoral research scholars, and faculty members. Further, its services may extend beyond the SRM University AP community, catering to academic and research institutions, universities, industries, and NGOs both within India and abroad, all on a fee-based basis.

Furthermore, CIF also serves as a venue for organizing short-term skill and training programmes focused on the utilization and application of various instruments and analytical techniques.

CIF: Central Instrumentation Facility

Faculty In-charge: The faculty in charge of the existing research instrument installed at the SRM University AP campus/PI of the proposed research facility under CIF.

Category-I User: SRM-AP faculty members, students, and staff working in Sponsored Research and Industrial Consultancy projects, academic research, and collaborators with other Institutes/Universities/Research organizations under MOU.

Category-II User: Faculty/Scientists, Students, and Staff of other Institutions/Universities/ Research organizations

Category-III User: Staff/Employees of Industries and Commercial Establishments

Objectives:

1. *Facilitating Cutting-edge Instrumentation Facilities:* Support state-of-the-art instrumentation resources to expedite both fundamental, applied, and advanced research endeavours by offering sample analysis services to researchers from both SRM University AP and external organizations, delivering dependable outcomes. Providing expert guidance in data interpretation, particularly in handling challenging data.
2. *Guidance and Training:* Provide guidance and training to individuals on the acquisition of data, operation, and maintenance of sophisticated instruments, ensuring their proficient utilization. Creating opportunities for networking among researchers in the realm of specific instrumentation, promoting synergistic growth in the field.
3. *Knowledge Dissemination:* Organize hands-on skill and training programmes, seminars, conferences, and symposia in collaboration with industrial and government partners, focused on specific instrumentation.

Standard Operating Procedures

Declaration of Instruments under CIF

Central Instrumentation Facility (CIF) at SRM University AP acquires research equipment/instruments considering the facility's research goals, needs assessment, and research prioritization considering the university's research strengths and focus areas. CIF invites proposals from the faculty in charge of the existing operational research instruments to register the instrument under central instrumentation facilities.

To develop new research facilities under CIF, faculty members need to submit a proposal against the university's "Central Instrumentation Facility" call.

1. *User Fees*: CIF will charge user fees to researchers, or projects that use their instrument facilities. These fees will contribute to revenue generation to support the ongoing maintenance of the instruments. The fees are based on an hourly rate, usage volume, or other factors depending on the research instrument. CIF also develops a clear strategy of instrumentation facility slot allocations to different research groups and projects to maximize their utilization.
2. *Annual Maintenance Budget*: A portion of the funds generated from user fees is typically allocated to the Annual Maintenance Budget. This budget is specifically designated for the routine maintenance, calibration, and repair of the instruments. The Dean Research

and Chief Finance and Accounts Officer (CFAO) will be responsible for managing this budget.

3. *Instrument Procurements*: Following the university's strengths, research goals, and areas of expertise, CIF will accept proposals from departments, research centers, and individual faculty members to procure the latest developed emerging trends and cutting-edge research instruments that best meet the research needs.
4. *Grants and Institutional Support*: Faculty-in-charge of individual instrument facilities may seek grants from government agencies, private foundations, or industry partners to support instrument operations and maintenance. Institutional support in the form of budget allocations from the university also contributes to the maintenance fund.
5. *Long-Term Planning*: CIFs will be engaged in long-term planning to ensure the sustainability of their instrument maintenance, upgrades, and replacements. This involves creating a reserve fund or endowment to handle major instrument repairs or replacements in the future.
6. *Regular Maintenance Contracts*: For Sophisticated Analytical Instruments, CIF will enter maintenance contracts with manufacturers or third-party service providers. This contract provides regular maintenance services for a fixed fee, to stabilize the annual maintenance charge.
7. *Technical Staff Recruitment*: Recruiting technical staff for a facility under CIF involves a university HR recruitment process to ensure that qualified and skilled individuals are hired to operate and maintain the instruments effectively. The Dean Research office will oversee the recruitment.
8. *Usage Tracking*: CIF will track the instrument usage and associated operational costs. These data adjust user fee structures and annual maintenance charges to reflect actual usage and maintenance needs.
9. *Documentation and Regular Review*: CIF will maintain a detailed record of instrument acquisitions, warranties, maintenance schedules, usage logs, and user feedback. CIF will also review instrument portfolio and consider upgrades or replacements as technology evolves or research needs change. CIF may solicit feedback from users to assess the performance of the instruments and the quality of CIF services for continuous improvement and accreditation purposes.
10. *Analytical Testing Audit and Accreditation*: Internal audits involving regular calibration and maintenance of equipment, proficiency testing, and inter-laboratory comparisons, help CIF to establish robust quality control measures to monitor and verify the accuracy and precision of analytical results. CIF may also be involved in the analytical testing audit and accreditation process following standardized procedures and specific quality standards of measurements to ensure the quality, accuracy, and reliability of analytical data generated by the facilities.

Procedure for Sample Collection and User Charges Payment:

For internal (Category-I) user, samples the student/ faculty member must fill up the specific form for the instrument, which is then reviewed by the respective faculty in charge during the submission of the samples. No direct user charge payment for category users; charges will be on record for internal report purposes. Yearly charges will be shared with the respective user (faculty members/research center) and will be collected from various internal fund sources (such as department funding, individual Professional Development Allowance, Professional Development Fund, and Professional Development Grant budget of the concerned faculty/students or project fund) as per the user approval.

For external (Category-II and III) users, there will be CGST and SGST as per Govt of India rules with the analysis charges and the payment must be made in the form of demand draft (DD), online transfer, or UPI payment gateway. For the Invoice, the user is requested to send/bring a self-addressed envelope with a stamp of the requisite amount.

Receipt/Invoice Format will be signed and released from the Dean research office – undersigned by the Coordinator, Central Research Facility & I-STEM

Policy for Internal Users (Category-I)

- Operational Personnel: Trained PhD students or technicians can operate the instruments for internal users.
- Mandatory Online Booking: Internal users must book instruments online through the booking system.
- Slots will be allotted to internal users on a first-come-first basis. However, for multiple slot requests by any user, the slots will be allotted based on the discretion of CIF considering the availability of the instruments.
- Presence Required: Internal users must be present during instrument operations.
- Pre-Payment: Pre-payment is not necessary for internal users as well. Charges will be on record and yearly collected from various internal fund sources (such as department funding, individual PDA, PDF, and PDG budget of the concerned faculty/students or project fund) as per the user approval. Annual fees can also be waived with the Vice-Chancellor approval.
- Free Slots for Ph.D. Students: Ph.D. students operating instruments for external user samples will receive complimentary slots for their research measurements.

Policy for External Users (Category-II and Category-III)

- Reserved Time Slots: Specific time slots per week are allocated for each instrument.
- Confirmation by CIF Admin: CIF Admin confirms the availability of slots to external users upon successful payment.

- Mandatory Pre-Payment: External users are required to make pre-payments before using the instruments.
- Operational Personnel: Only technicians or Ph.D. students from SRMAP are permitted to operate instruments for external users.
- Presence during Operation: External users may or may not need to be present during instrument operation.
- Raw data in the form of analysis reports will be sent by email or the user may pick them up at the reception of CIF. All the records will be stored in CIF deposits for a maximum of one year following analysis.
- Specifications of instrumental conditions utilized in the analysis as well as calibration curves, calculated concentrations, and matching library data may be provided.
- CIF will ensure to safeguard the data privacy and strictly preserve the IPR of the users.

User Fee Payment Process

Payment of the user charges can be deposited through any of the modes of payment:

(i) Through Online transactions

Account No.	50100663679048
Account Holder Name	SRM University AP Unit of SRM TRUST AP
Bank Name	HDFC Bank Ltd.
Branch Name and code	Mangalagiri, 2704
IFSC/MICR code	HDFC0002704/520240302
* Please mention your full name during online payment	

(ii) Through Demand Draft:

The demand Draft should be in favour of SRM University AP Unit of SRM TRUST AP payable at Mangalagiri.

Terms and Conditions

1. All publications of research work, where the analytical services of the CIF, SRM University have been made use of, shall be duly acknowledged (e.g., *The services extended by the Central Instrumentation Facility, SRM University-AP, Andhra Pradesh in the XRD analysis are duly acknowledged*).
2. The content of our report should not be used for any advertisement, evidence, litigation, or/and quote as a certificate to a third party.

Administrative Responsibilities

- Develop and periodic update of the Central Instrumentation Facility Webpage and User Registration portal with the Communication team.
- Comprehensive documentation for every equipment by the Research Office, subject to biannual operational audits
- Maintenance of Payment gateway to accommodate both internal and external users.
- Coordinating procurement activities, and recruitment of technicians with the respective faculty *i/c* of each instrumentation facility.
- Aiding the faculty *i/c* in managing repairs and maintenance tasks.
- Offering additional support to the faculty *i/c* in handling Annual Maintenance Contracts (AMC) with vendors.
- Conducting training for a minimum of two PhD students per instrument, either from the relevant department or based on user needs.

Fund Allocations

- The faculty in-charge, along with Dean Research /CRF Coordinator will define instrument usage fee charges for both external and internal users of each research facility as per the nationalized rate. This may be revised from time to time.
- For the initial sustainability of CIF, a dedicated fund will be initiated using the research budget. This fund will be available to support instrument repair, maintenance, and AMC needs during the initial two years of CIF establishment.
- The user fee proceeds will be earmarked for instrument repair, maintenance, and AMC expenses from the overhead charges as per the university policy.

A list of instruments available at CIF and their scope is provided in **Annexure-I**.

The details of usage fee charges are given in **Annexure II**.

Sample requisition forms for individual instruments are given in **Annexure III**.

Registrar

Dr. R. PREMKUMAR
REGISTRAR
SRM UNIVERSITY - AP
Andhra Pradesh - 522240

Annexure-I: List of Equipment under Central Instrumentation Facility

Sl No.	Name of the Equipment	Description
1	**X-ray Diffractometer**	Automated Multipurpose Powder X-ray diffractometer for crystal structure and defect measurements of various materials
2	**ICEP-OES**	Inductively Coupled Plasma Optical Emission spectroscopy is an analytical technique to determine how much of certain chemical elements are present in a sample
3	**Sputtering cum Thermal evaporator**	Growth tool for Metallic and semiconducting thin films
4	**Low-Temperature Electrical Probe Station with Source Measurement Unit and Impedance Analyser**	The instrument supports current-voltage and electrical impedance characterizations and testing of semiconductors, optoelectronics devices, and materials within a temperature limit of liquid nitrogen (77K) to 350K

Annexure-II: Proposed Instrument Usage Charges (in INR)

Sl No.	Name of the Equipment	Category-I: SRM AP faculty, students, and staff	Category-II: Other Universities/Research Organizations*	Category-III: Charges for Industry*
1	X-ray Diffractometer	150 / hr	300 / hr	500 / hr
2	ICEP-OES Solid- prepared sample	100/Sample	500/Sample	3000/Sample
	ICEP-OES Solid- unprepared sample	600/Sample	1500/Sample	5000/Sample
	ICEP-OES Liquid- Prepared sample	100/Sample	500/Sample	3000/Sample
	ICEP-OES Liquid- unprepared sample	150/Sample	600/Sample	3500/Sample
3	Sputtering cum Thermal evaporator	200/Sample	400/Sample	800/Sample
4	Low-Temperature Electrical Probe Station with Source Measurement Unit and Impedance Analyser	RT – 100/Sample LN2 – 200/Sample	RT – 200/Sample LN2 – 400/Sample	RT – 500/Sample LN2 – 1000/Sample

* Cost of Consumables will be extra

#User needs to arrange deposition materials/metal targets, else additional charges depending on material/metal target cost

RT – Room temperature electrical measurements

LN2 - Electrical measurements with liquid nitrogen

Annexure-III: Sample Submission Template

XRD SAMPLE SUBMISSION FORM

(Please complete and submit this form along with samples. Leave the boxes blank if not applicable to you)

Name of the User			
Supervisor			
Affiliation			
Type of the samples	**Powder/pellet/film**	**Scan Time (h) (typical 0.5 h)**	
2θ range		**Steps size**	
Scan speed		**Measurement Mode**	
Chemical Composition (elements present)			
No of Samples			
Hazard/Risk Info			
E-mail/Phone			

For Internal/Office Use only

Reference No.	
Samples Received Date	
Completion Date	
Total scan time (h)	
Charges incurred	
Budget Head	
Authorized Signature	

ICP-OES SAMPLE SUBMISSION FORM

(Please complete and submit this form along with samples. Leave the boxes blank if not applicable for you)

Name of the User	
Supervisor	
Affiliation	
Hazard/Risk Info	
E-mail/Phone	

To be filled by the Applicant										**To be filled by Lab Supervising Faculty**	
Sl. No	**Sample type**	**Color**	**Nature of the sample**	**Volatility**	**Visible Turbidity/ pptn.**	**pH**	**TDS (mg/ L)**	**Cond. (mS)**	**Filt. (by 0.2 mm)**	**Dilution Fac.**	**Other Obs. /Comments**
1	Choose an item.	Choose an item.	Choose an item.	Choose an item.	Choose an item.				Choose an item.		
2	Choose an item.	Choose an item.	Choose an item.	Choose an item.	Choose an item.				Choose an item.		
3	Choose an item.	Choose an item.	Choose an item.	Choose an item.	Choose an item.				Choose an item.		
4											
5											

For Internal/Office Use only

Reference No.		**Total scan time (h)**	
Samples Received Date		**Charges incurred**	
Completion Date		**Budget Head**	
Authorized Signature			

SPUTTERING THERMAL EVAPORATOR
SAMPLE SUBMISSION FORM

(Please complete and submit this form along with samples. Leave the boxes blank if not applicable for you)

Name of the User			
Supervisor			
Affiliation			
Type of the Deposition	Choose an item.	Deposition Rate (Å/Sec)	
Deposition material		Vacuum Pressure	
Substrate details			
No of Samples			
Hazard/Risk Info			
E-mail/Phone			

For Internal/Office Use only

Reference No.	
Samples Received Date	
Completion Date	
Total scan time (h)	
Charges incurred	
Budget Head	
Authorized Signature	

PROBE STATION ELECTRICAL MEASUREMENTS
SAMPLE SUBMISSION FORM

(Please complete and submit this form along with samples. Leave the boxes blank if not applicable for you)

Name of the User			
Supervisor			
Affiliation			
Measurements Mode	Choose an item.	**Measurement Temperature range**	Choose an item.
Chemical Composition (elements present)			
No of Samples			
Hazard/Risk Info			
E-mail/Phone			

For Internal/Office Use only

Reference No.	
Samples Received Date	
Completion Date	
Total scan time (h)	
Charges incurred	
Budget Head	
Authorized Signature	

Dr. R. PREMKUMAR
REGISTRAR
SRM UNIVERSITY - AP
Andhra Pradesh - 522240